

EC 441 - Week 10 Problem Set

Transport Layer, UDP, and Reliable Data Transfer

Name: _____ Date: _____

Instructions. Show your work for calculation problems. For conceptual problems, answer in complete sentences and use networking vocabulary precisely.

Problem 1. Transport Layer Purpose

In one or two sentences, explain what the transport layer provides that the network layer does not.

Problem 2. Multiplexing and Demultiplexing

A host is running a web browser, an SSH client, and a video call at the same time. Explain how the transport layer knows which incoming data belongs to which application.

Problem 3. UDP Header

A UDP segment carries 250 bytes of application data.

- (a) What is the size of the UDP header?
- (b) What value should be placed in the UDP length field?

Problem 4. UDP vs. TCP

Give two examples of applications or protocols where UDP is a good choice. For each one, briefly explain why UDP makes sense.

Problem 5. Transmission Time

A packet is 1500 bytes long and is sent over a 100 Mb/s link. Calculate the transmission time in seconds and microseconds.

Problem 6. Stop-and-Wait Utilization

A sender uses Stop-and-Wait on a link with the following values:

$B = 100 \text{ Mb/s}$, $RTT = 80 \text{ ms}$, packet size = 1500 bytes

Calculate the Stop-and-Wait utilization.

Problem 7. Useful Throughput

Using the utilization from Problem 6, calculate the useful throughput in bits per second.

Problem 8. Bandwidth-Delay Product

A network path has the following values:

$B = 500 \text{ Mb/s}$, $RTT = 100 \text{ ms}$

Calculate the bandwidth-delay product in bits and megabits.

Problem 9. Required Window Size

Using the path from Problem 8, assume each packet is 1000 bytes. How many packets must be in flight to fully utilize the link?

Problem 10. Why Sequence Numbers Matter

In Stop-and-Wait, suppose a data packet arrives correctly, but the ACK is lost. The sender times out and retransmits the same packet. Explain why sequence numbers are necessary in this situation.

Problem 11. Go-Back-N Packet Loss

A Go-Back-N sender has a window size of 4 and sends packets 0, 1, 2, and 3. Packet 1 is lost, but packets 2 and 3 arrive at the receiver.

Explain what the receiver does with packets 2 and 3. Then explain what the sender retransmits after the timer expires.

Problem 12. Selective Repeat Packet Loss

A Selective Repeat sender has a window size of 4 and sends packets 0, 1, 2, and 3. Packet 1 is lost, but packets 2 and 3 arrive at the receiver.

Explain what the receiver does with packets 2 and 3. Then explain what the sender retransmits after the timer for packet 1 expires.

Problem 13. Sequence Number Constraint for Go-Back-N

A Go-Back-N protocol uses 4-bit sequence numbers. What is the maximum allowed window size?

Problem 14. Sequence Number Constraint for Selective Repeat

A Selective Repeat protocol uses 4-bit sequence numbers. What is the maximum allowed window size?

Problem 15. TCP Connection to GBN and SR

TCP is not exactly Go-Back-N or Selective Repeat. Explain one idea TCP takes from Go-Back-N and one idea TCP takes from Selective Repeat.

